

Radiative Cooling on Coarse-Grained TRML Fields

1 Setup and notation

The fine simulation is a 2D turbulent radiative mixing layer (TRML) at 512×1024 on a 10×20 pc box, following the setup of Sharma et al. (2025) but in 2D. The fine cell size is $\delta x = 10 \text{ pc}/512 = 0.0195$ pc. Coarse-graining by a factor Δ (in cells) gives coarse cells of physical size $\ell = \Delta \delta x$; $\Delta = 32$ gives a 16×32 grid with $\ell = 0.625$ pc, and $\Delta = 64$ gives 8×16 with $\ell = 1.25$ pc. In 2D, “volume” means area throughout.

1.1 Block mean

For a field f ,

$$\mathcal{B}[f]_{ij} \equiv \text{BlockMean}(f, \Delta) = \frac{1}{\Delta^2} \sum_{n=0}^{\Delta-1} \sum_{m=0}^{\Delta-1} f(i\Delta + n, j\Delta + m).$$

$\mathcal{B}$ is linear and conserves integrals: $\sum_{ij} \mathcal{B}[f]_{ij} \ell^2 = \int f dA$.

1.2 Coarse fields

Conserved and extensive quantities are volume averaged; specific quantities are mass (Favre) averaged:

$$\bar{\rho} = \mathcal{B}[\rho], \quad \bar{u}_i = \mathcal{B}[\rho u_i]/\bar{\rho}, \quad \bar{P} = \mathcal{B}[P], \quad \bar{s} = \mathcal{B}[\rho s]/\bar{\rho}.$$

The internal energy density $P/(\gamma - 1)$ is extensive, so volume averaging it is correct, but it is not separately conserved. The coarse state therefore satisfies $\mathcal{B}[E] = \bar{E}_{\text{int}} + \frac{1}{2} \bar{\rho} |\bar{u}|^2 + k_{\text{sgs}}$, with subgrid kinetic energy $k_{\text{sgs}} = \frac{1}{2} (\mathcal{B}[\rho |u|^2] - \bar{\rho} |\bar{u}|^2) \geq 0$. Averaging P directly keeps k_{sgs} out of the temperature; reconstructing P from $\mathcal{B}[E]$ would raise T_{cg} by $(\gamma - 1) \mu m_p k_{\text{sgs}} / (k_B \bar{\rho})$.

1.3 Coarse temperature

From the equation of state applied to coarse fields,

$$T_{\text{cg}} = \frac{\mu m_p \bar{P}}{k_B \bar{\rho}} = \frac{\mathcal{B}[\rho T]}{\mathcal{B}[\rho]},$$

i.e. T_{cg} is the Favre (mass-weighted) temperature, biased toward the cold phase. By contrast $\mathcal{B}[T]$ is volume weighted and biased toward the hot phase. If the block is isobaric, T_{cg} is exactly the harmonic volume mean,

$$T_{\text{cg}}^{-1} = \mathcal{B}[T^{-1}] \quad (P = \text{const in block}). \quad (1)$$

2 Cooling Functions

The cooling function for a high resolution field can be written as

$$\mathcal{C} = n^2 \Lambda(T)$$

We define the true cooling on the low-resolution field as

$$\mathcal{C}_{\text{true}} = \mathcal{B}[n^2 \Lambda(T)]$$

However, working with the low resolution fields we may lose information. The resolved cooling after coarse graining, $\mathcal{C}_{\text{cg}}$ is defined as

$$\mathcal{C}_{\text{cg}} = \bar{n}^2 \Lambda(\bar{T})$$

2.1 Uniform Density Estimator

Assume each cell has uniform density. Then we can take n^2 out as constants, giving us $\bar{n}^2 \mathcal{B}[\Lambda(T)]$. Given the volume weighted pdf $\mathcal{P}_V(T)$ for each cell, we can write

$$\mathcal{B}[\Lambda(T)] = \int dT \Lambda(T) \mathcal{P}_V(T) = \Lambda_{\text{PDF}}$$

We define the Uniform Density Cooling Estimator as

$$\mathcal{C}_{\text{n}} = \bar{n}^2 \Lambda_{\text{PDF}}$$

2.2 Uniform Pressure Estimator

Assume each cell has uniform pressure $\bar{P}$. We rewrite $n = P/k_B T$.

$$\mathcal{B}[n^2 \Lambda(T)] = \left(\bar{P}/k_B\right)^2 \mathcal{B}[\Lambda(T)/T^2]$$

We can estimate the $\mathcal{B}[\Lambda(T)/T^2]$ with a volume weighted PDF, given by

$$\mathcal{B}[\Lambda(T)/T^2] = \int dT \mathcal{P}_V(T) \frac{\Lambda(T)}{T^2}$$

The $\mathcal{C}_{\text{P}}$ can be defined as

$$\mathcal{C}_{\text{P}} = \left(\frac{\bar{P}}{k_B}\right)^2 \int dT \mathcal{P}_V(T) \frac{\Lambda(T)}{T^2}$$

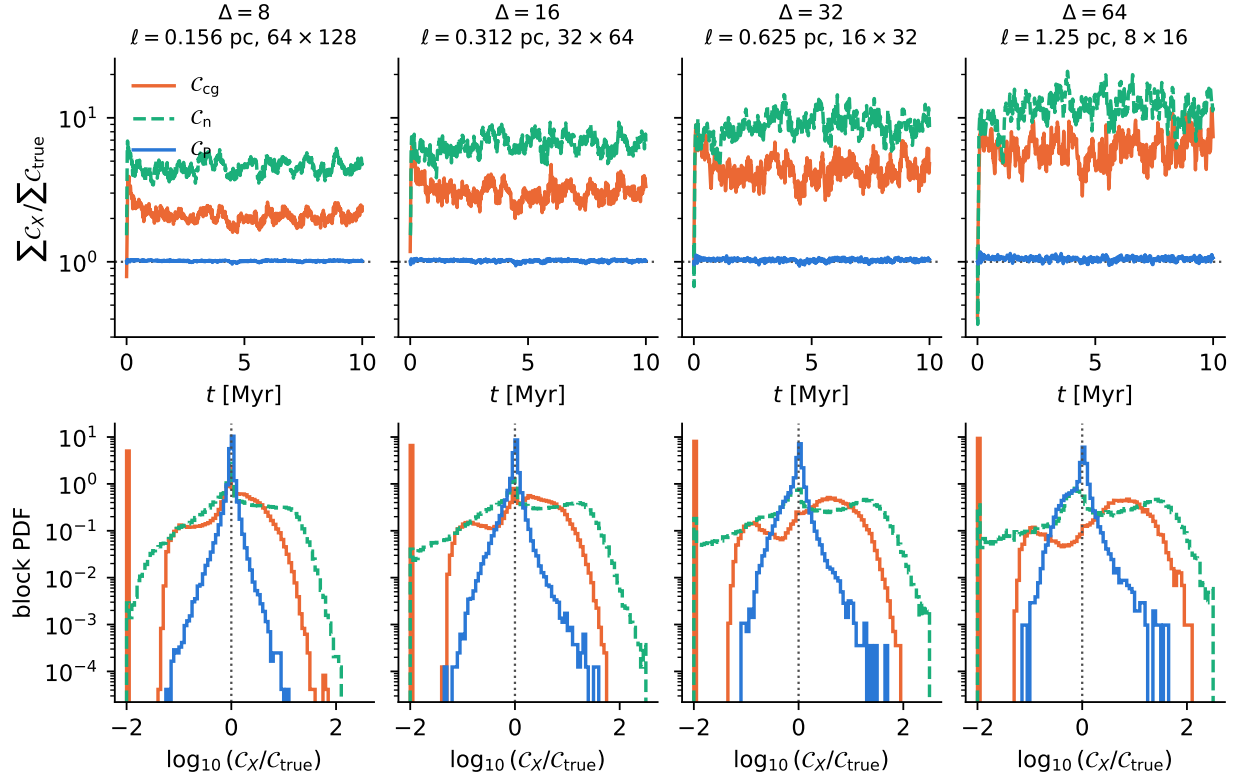


Figure 1: The three cooling estimates against $\mathcal{C}_{\text{true}} = \mathcal{B}[n^2 \Lambda(T)]$ at four coarse-graining factors. Top: box-integrated cooling relative to the truth, against time. Bottom: distribution of the block-by-block error over blocks with $\mathcal{C}_{\text{true}} > 0$ and $t \geq 5$ Myr; values are clipped to the axis range, so the spike at the left edge counts blocks where the estimate is zero (for $\mathcal{C}_{\text{cg}}$, T_{cg} falls outside the cooling window).

2.3 Comparing Cooling

We evaluate $\mathcal{C}_{\text{true}}$, $\mathcal{C}_{\text{cg}}$, $\mathcal{C}_n$ and $\mathcal{C}_p$ on every snapshot of the 512×1024 run for $\Delta = 8, 16, 32, 64$. $\mathcal{C}_n$ and $\mathcal{C}_p$ use the unbinned volume PDF of the fine cells, so the PDF integrals reduce to block means, $\mathcal{B}[\Lambda(T)]$ and $\mathcal{B}[\Lambda(T)/T^2]$. n , T and Λ follow AthenaK exactly: $n = \rho/(\mu m_u)$, $T = \mu m_u P/(k_B \rho)$, and ISMCoolFn with cooling switched off for $T \leq 1.05 \times 10^4$ K and $T > 0.95 \times 10^6$ K.

Figure 1 shows that $\mathcal{C}_p$ is the only estimator that recovers the box-integrated cooling: it stays within a few per cent of $\mathcal{C}_{\text{true}}$ at every Δ (about 1% at $\Delta = 8$, rising to about 5% at $\Delta = 64$), and its block errors are sharply peaked at zero. $\mathcal{C}_{\text{cg}}$ overestimates the total by a factor of about 2 at $\Delta = 8$ and about 6 at $\Delta = 64$, because T_{cg} is pulled into the mixing-layer peak of Λ while the fine gas sits mostly in the non-cooling phases; it also returns zero in 25% (at $\Delta = 8$) to 47% (at $\Delta = 64$) of the blocks that do cool, because their T_{cg} lands outside the cooling window. $\mathcal{C}_n$ is worst, about 5 to 12 times too large: at fixed $\bar{n}$ it gives the hot, dilute gas the density of the cold phase, and the error grows with Δ as blocks mix more of both phases. Both failures come from the density weighting. Because the mixing layer is close to isobaric, $n \propto 1/T$ inside a block, and only $\mathcal{C}_p$ carries that factor through $\Lambda(T)/T^2$.

3 Subgrid Kinetic Energy

The conserved quantity is the total energy, given by

$$E = \frac{1}{2} \rho |u|^2 + P/(\gamma - 1)$$

The coarse solver would have access to $\mathcal{B}[E]$, not necessarily $\mathcal{B}[P]$. We could still volume average and get $\bar{P} = \mathcal{B}[P]$.

We also have to deal with unresolved Kinetic Energy, k_{sgs} .

$$k_{\text{sgs}} \equiv \frac{1}{2} \left(\mathcal{B}[\rho |u|^2] - \bar{\rho} |\bar{u}|^2 \right)$$

The energy is now decomposed as

$$\mathcal{B}[E] = \frac{\mathcal{B}[P]}{\gamma - 1} + k_{\text{sgs}} + \frac{|\mathcal{B}[\rho u]|^2}{2\bar{\rho}}$$

Let's assume isobaricity, giving us

$$\bar{\rho} = \frac{\mu m_p}{k_B} \bar{P} \int dT \mathcal{P}_V(T) \frac{1}{T}$$

For convinience let's define the "Harmonic Temperature" T_H as

$$T_H \equiv \left(\int \mathcal{P}_V(T) T^{-1} dT \right)^{-1}$$

Substituting this into the energy decomposition, we get:

$$k_{\text{sgs}} = \mathcal{B}[E] - \frac{|\mathcal{B}[\rho u]|^2}{2\bar{\rho}} - \frac{k_B \bar{\rho} T_H}{(\gamma - 1) \mu m_p}$$

Let’s say all the unresolved kinetic energy goes into heat, and compute a “naive temperature”, from the resolved quantities.

$$T_{\text{naive}} = \frac{(\gamma - 1)\mu m_p}{k_B \bar{\rho}} \left(\mathcal{B}[E] - \frac{1}{2} \bar{\rho} |\bar{\mathbf{u}}|^2 \right)$$

Now we can write the subgrid kinetic energy as the excess of the derived Harmonic temperature and naive temperature.

$$k_{\text{sgs}} = \frac{k_B \bar{\rho}}{(\gamma - 1)\mu m_p} (T_{\text{naive}} - T_H)$$

3.1 Testing the isobaric closure

The closure is exact up to the isobaric step. Substituting $\mathcal{B}[E] - \frac{1}{2} \bar{\rho} |\bar{\mathbf{u}}|^2 = \bar{P}/(\gamma - 1) + k_{\text{sgs}}$ into T_{naive} gives

$$k_{\text{sgs}}^{\text{model}} - k_{\text{sgs}} = \frac{\bar{P} - P_H}{\gamma - 1}, \quad P_H \equiv \frac{k_B \bar{\rho} T_H}{\mu m_p} = \frac{\mathcal{B}[\rho]}{\mathcal{B}[\rho/P]}, \quad (2)$$

so the model error is the gap between the volume-mean pressure and the pressure an isobaric block with the same $\bar{\rho}$ and T_H would have. We compare the closure with the definition on the same 512×1024 run and the same Δ , with $T_H = \mathcal{B}[T^{-1}]^{-1}$ taken from the unbinned volume PDF of the fine cells.

Figure 2 shows that the closure tracks the time evolution and the growth with Δ of the box-mean k_{sgs} , but overestimates it by 31–36% at every Δ . Block by block the median ratio $k_{\text{sgs}}^{\text{model}}/k_{\text{sgs}}$ is about 1.5, and roughly 2% of blocks give $k_{\text{sgs}}^{\text{model}} < 0$. The error is set by Equation 2, not by any approximation in the kinetic terms. k_{sgs} is only 0.3% ($\Delta = 8$) to 3% ($\Delta = 64$) of $\bar{P}/(\gamma - 1)$, so relative pressure departures $(\bar{P} - P_H)/\bar{P}$ of the same order, well within what the turbulence and cooling in the mixing layer produce, are enough to swamp it. The same effect sets the floor at small k_{sgs} in the bottom panels: in quiet blocks the residual pressure fluctuation, not the velocity dispersion, dominates $T_{\text{naive}} - T_H$.

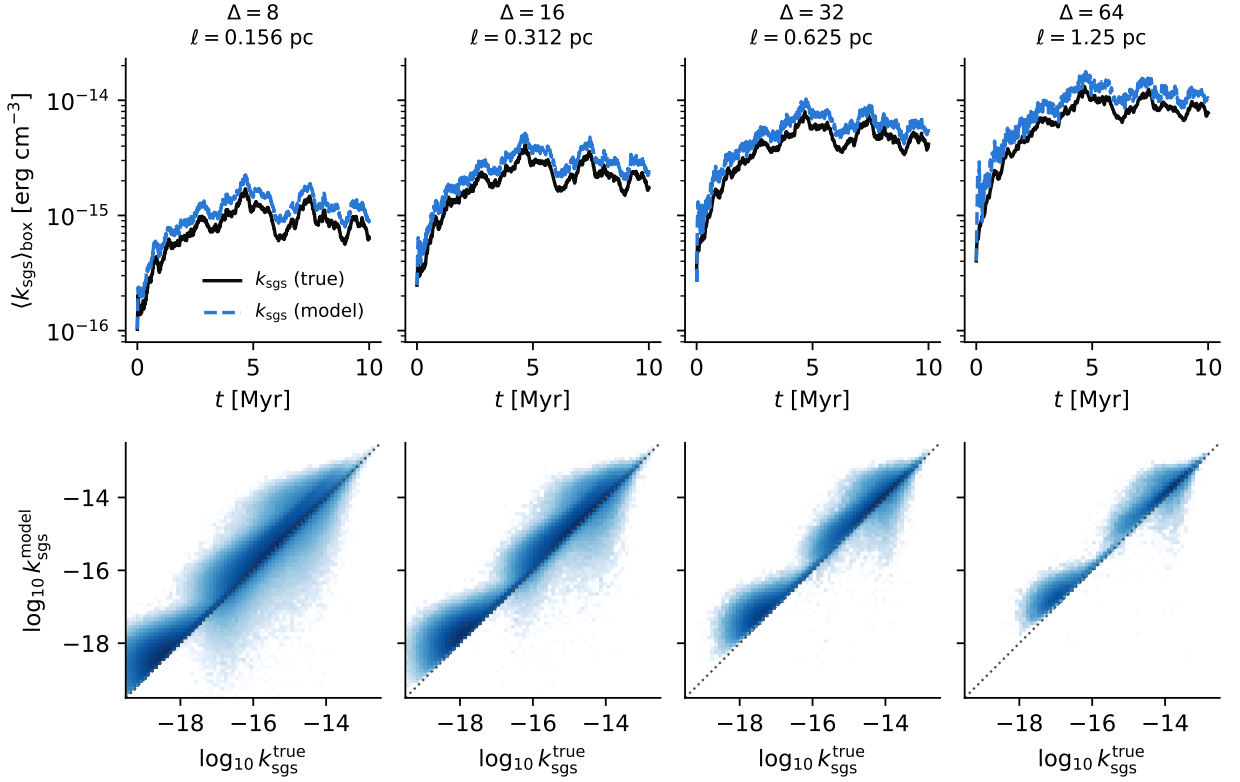


Figure 2: The isobaric closure $k_{\text{sgs}}^{\text{model}} = k_B \bar{\rho} (T_{\text{naive}} - T_H) / [(\gamma - 1) \mu m_p]$ against the definition $k_{\text{sgs}} = \frac{1}{2} (\mathcal{B}[\rho |u|^2] - \bar{\rho} |\bar{u}|^2)$. Top: box-mean k_{sgs} against time. Bottom: block-by-block joint distribution for $t \geq 5$ Myr, in erg cm^{-3} ; the dotted line is equality. The 2% of blocks with $k_{\text{sgs}}^{\text{model}} \leq 0$ are not shown.